
Creation of a Digital Forensic Examination System with Mobile Application Artifact Analysis

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ABSTRACT

Thesis Title: Development of a Digital Forensic Examination System for Mobile Applications to Identify Security Artifacts and Analyze Incidents

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English Version:

This diploma project presents the development and implementation of a digital forensic examination system for forensic applications designed to identify security artifacts, automate forensic analysis, and support cybersecurity incident investigations in modern digital environments.

Forensic applications increasingly process and store sensitive personal, authentication, and corporate data, making them attractive targets for malware activity, unauthorized access, credential theft, data leakage, and other cybersecurity threats. Traditional forensic investigation approaches often focus primarily on full-device acquisition and manual evidence analysis, which significantly increases investigation complexity, processing time, and the possibility of human error. In addition, many existing forensic tools provide limited automation capabilities and insufficient support for application-level artifact analysis and correlation.

The proposed system addresses these limitations through the implementation of an automated forensic analysis platform focused specifically on desktop forensic application artifacts and suspicious behavioral indicators. The developed solution performs structured scanning of application-related data sources, including log files, cached content, authentication traces, session information, network indicators, temporary files, and suspicious activity records. The system applies rule-based artifact detection methods combined with a quantitative risk scoring model to evaluate the severity of identified findings and prioritize potential security threats.

The implementation was developed using Python and includes several integrated modules responsible for artifact acquisition, anomaly detection, forensic correlation, risk evaluation, statistical analysis, and automated report generation. The system supports identification of suspicious IP addresses, authentication tokens, traces of file deletion, abnormal session activity, suspicious network communication indicators, and other forensic artifacts associated with potential compromise or malicious behavior. A graphical user interface was developed to simplify user interaction and provide accessible forensic analysis functionality for both technical and non-technical investigators.

The developed platform automatically generates structured forensic reports containing

detected anomalies, severity classifications, metadata, integrity-related information, and analytical summaries. In addition, the system provides graphical visualization of risk distribution and statistical analysis results, improving readability and supporting forensic decision-making processes during cybersecurity investigations.

The research also includes an analysis of existing mobile forensic approaches, current challenges in digital evidence processing, machine learning applications in cybersecurity analysis, and limitations of traditional forensic solutions. Experimental evaluation shows that the proposed system improves investigation efficiency, reduces manual analysis complexity, automates evidence processing procedures, and provides consistent detection of predefined forensic artifacts across multiple test datasets and simulated incident scenarios.

The practical significance of this diploma project lies in supporting cybersecurity specialists, incident response teams, and digital forensic investigators during desktop forensic application security investigations through automated artifact analysis, structured evidence processing, and reproducible forensic reporting workflows.

Keywords: Digital Forensics, Mobile Security, Incident Analysis, Security Artifacts, Forensic applications, Risk Scoring, Cybersecurity, Artifact Detection, Forensic Analysis, Evidence Processing, Python, Digital Investigation, Automated Reporting, Threat Detection.

Қазақша нұсқасы:

Бұл дипломдық жоба мобильді қосымшалардағы қауіпсіздік артефактілерін анықтауға, цифрлық дәлелдерді автоматтандырылған түрде талдауға және киберқауіпсіздік инциденттерін зерттеуге арналған цифрлық криминалистикалық сараптама жүйесін әзірлеуге бағытталған.

Қазіргі уақытта мобильді қосымшалар пайдаланушылардың жеке деректерін, аутентификация ақпаратын және корпоративтік мәліметтерді кең көлемде өңдейді және сақтайды. Осыған байланысты олар зиянды бағдарламалардың, рұқсатсыз қол жеткізудің, деректердің ағып кетуінің және басқа да киберқауіптердің негізгі нысанасына айналууда. Дәстүрлі цифрлық криминалистика тәсілдері көбінесе құрылғыны толық зерттеуге және қолмен талдау жүргізуге негізделген, бұл зерттеу уақытын ұзартып, процестің күрделілігін арттырады.

Ұсынылған жүйе осы мәселелерді шешу үшін мобильді қосымшалар артефактілерін автоматтандырылған түрде талдайтын арнайы платформаны ұсынады. Жүйе журнал файлдарын, кэш деректерін, сессия ақпаратын, аутентификация іздерін, желілік индикаторларды және күмәнді әрекеттерді сканерлейді және талдайды. Сонымен қатар жүйе қауіп деңгейін бағалау үшін ережеге негізделген артефакт анықтау әдістері мен тәуекелді бағалау моделін қолданады.

Бағдарламалық шешім Python тілінде әзірленген және артефакттарды жинау, аномалияларды анықтау, тәуекелдерді бағалау, статистикалық талдау және автоматтандырылған есеп генерациясы модульдерін қамтиды. Жүйе күмәнді IP-мекенжайларды, аутентификация токендерін, файлдарды жою іздерін, күмәнді сессия белсенділігін және басқа да қауіпсіздік индикаторларын анықтай алады.

Сонымен қатар жүйе анықталған артефакттар мен қауіп деңгейлері туралы құрылымдалған криминалистикалық есептерді автоматты түрде қалыптастырады. Тәуекел көрсеткіштерінің визуализациясы мен статистикалық графиктері зерттеу нәтижелерін түсінуді жеңілдетеді және киберқауіпсіздік саласындағы шешім қабылдау процестерін қолдайды.

Зерттеу барысында мобильді криминалистика әдістері, цифрлық дәлелдерді өңдеу мәселелері және дәстүрлі криминалистикалық құралдардың шектеулері талданды. Тәжірибелік бағалау нәтижелері ұсынылған жүйенің тергеу тиімділігін арттыратынын, қолмен орындалатын талдау көлемін азайтатынын және криминалистикалық зерттеу процестерін автоматтандыратынын көрсетті.

Түйін сөздер: Цифрлық криминалистика, Киберқауіпсіздік, Мобильді қауіпсіздік, Артефакттарды анықтау, Инциденттерді талдау, Дәлелдерді өңдеу, Python, Тәуекелдерді бағалау, Қауіптерді анықтау, Форензикалық талдау.

Русская версия:

Данный дипломный проект посвящен разработке и реализации системы цифровой криминалистической экспертизы мобильных приложений, предназначенной для выявления артефактов безопасности, автоматизации анализа цифровых доказательств и поддержки расследования киберинцидентов в современных цифровых средах.

Мобильные приложения активно обрабатывают и хранят персональные данные пользователей, аутентификационную информацию и корпоративные данные, что делает их привлекательной целью для вредоносного программного обеспечения, несанкционированного доступа, кражи учетных данных, утечек информации и других киберугроз. Традиционные подходы цифровой криминалистики в основном ориентированы на полный анализ устройства и ручную обработку доказательств, что значительно увеличивает сложность расследования, время анализа и вероятность человеческих ошибок. Кроме того, многие существующие криминалистические инструменты обладают ограниченными возможностями автоматизации и недостаточной поддержкой анализа артефактов на уровне приложений.

Предлагаемая система направлена на решение данных ограничений посредством разработки автоматизированной платформы анализа артефактов мобильных приложений и подозрительных поведенческих индикаторов. Разработанное решение выполняет структурированное сканирование связанных с приложениями источников данных, включая журналы событий, кэшированные файлы, следы аутентификации, информацию о сессиях, сетевые индикаторы, временные файлы и записи подозрительной активности. Система использует методы обнаружения артефактов на основе правил в сочетании с моделью количественной оценки риска для определения уровня опасности обнаруженных аномалий и приоритизации потенциальных угроз безопасности.

Программная реализация была разработана на языке Python и включает интегрированные модули для сбора артефактов, обнаружения аномалий, криминалистической корреляции, оценки рисков, статистического анализа и автоматизированного формирования отчетов. Система поддерживает выявление подозрительных IP-адресов, токенов аутентификации, следов удаления файлов, подозрительной активности сессий, сетевых индикаторов и других цифровых артефактов, связанных с потенциальной компрометацией или вредоносной активностью.

Кроме того, система автоматически формирует структурированные криминалистические отчеты, содержащие обнаруженные аномалии, уровни критичности, метаданные, информацию о целостности данных и аналитические сводки. Визуализация распределения рисков и статистических результатов

повышает наглядность анализа и поддерживает процессы принятия решений при проведении расследований в области кибербезопасности.

В ходе исследования были рассмотрены современные методы мобильной криминалистики, проблемы обработки цифровых доказательств, применение методов машинного обучения в анализе киберугроз, а также ограничения традиционных криминалистических решений. Экспериментальная оценка продемонстрировала, что предложенная система повышает эффективность расследований, снижает объем ручного анализа, автоматизирует обработку доказательств и обеспечивает стабильное выявление криминалистических артефактов в различных тестовых сценариях.

Практическая значимость дипломного проекта заключается в поддержке специалистов по кибербезопасности, команд реагирования на инциденты и цифровых криминалистов при расследовании инцидентов мобильных приложений посредством автоматизированного анализа артефактов, структурированной обработки доказательств и воспроизводимого формирования криминалистических отчетов.

Ключевые слова: Цифровая криминалистика, Кибербезопасность, Мобильная безопасность, Анализ инцидентов, Артефакты безопасности, Обработка доказательств, Python, Оценка рисков, Обнаружение угроз, Форензический анализ, Автоматизированная отчетность.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
APT	Advanced Persistent Threat
CLI	Command-Line Interface
CSV	Comma-Separated Values
DPI	Deep Packet Inspection
GUI	Graphical User Interface
IoC	Indicator of Compromise
JSON	JavaScript Object Notation
ML	Machine Learning
MITM	Man-in-the-Middle
MobSF	Mobile Security Framework
NIST	National Institute of Standards and Technology
PDF	Portable Document Format
PII	Personally Identifiable Information
SQLite	Structured Query Language Lite
STRIDE	Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privi- lege
UML	Unified Modelling Language

1. Introduction

1.1. Actuality of the Research

forensic applications are increasingly serving as repositories for sensitive personal and corporate data. In the event of a security incident, a forensic examination extending beyond the physical device to the applications themselves becomes imperative. This application-centric analysis is crucial for purposes such as reconstructing incident time-lines, extracting critical credentials or other digital artifacts, and effectively supporting both incident response and legal processes. However, current forensic methodologies often emphasize broad device imaging or selective artifact retrieval, thereby frequently lacking a truly integrated, application-focused framework. Such a framework would ideally unify techniques such as static analysis, dynamic instrumentation, memory forensics, and automated artifact correlation to provide a more holistic investigative capability.

1.2. Research Problem

There is a gap between rapidly-evolving mobile app ecosystems and forensic tooling: app-level encryption, obfuscation, ephemeral memory artifacts, and cloud integration complicate reproducible evidence extraction. Investigators need an extensible, documented system that prioritizes forensic soundness (hashing, chain of custody) while automating repetitive tasks to reduce human error and analysis time.

1.3. Objectives

The objectives of this research are:

1. analyze existing mobile forensic tools and research approaches;
2. identify key forensic artifacts generated by forensic applications;
3. design the architecture of a mobile forensic analysis system;
4. develop a desktop forensic application for digital forensic artifact analysis;
5. implement anomaly detection methods for security incidents;
6. evaluate the effectiveness of the proposed system.

2. Literature Review and Analysis of Current Research

2.1. Analysis of the Current State of Research

The proliferation of forensic applications and their capacity to manage substantial volumes of sensitive personal and corporate data have elevated mobile digital forensics to a critical area of research. In the event of a security incident, forensic analysis of these applications enables investigators to reconstruct user actions, identify security transgressions, and gather digital evidence for legal proceedings.

Recent studies demonstrate a growing shift from full device acquisition toward application-level forensic analysis. Researchers emphasize that forensic applications generate a wide range of forensic artifacts, including authentication tokens, log files, cached data, session information, and network traces [1, 2, 3]. These artifacts provide valuable insights into application behavior and potential security incidents.

International guidelines and standards, such as NIST SP 800-101, highlight the importance of structured and systematic forensic processes for mobile environments, including artifact identification, preservation, and analysis [4]. Overall, current research confirms the need for automated and application-focused forensic approaches.

2.2. Machine Learning Methods in Mobile Forensics

Machine learning techniques are increasingly applied in mobile forensics to improve the detection of malicious behavior and anomalous application activity. Traditional rule-based approaches are effective for identifying known patterns but often fail to detect new or previously unseen threats [8].

Recent studies show that ensemble-based machine learning models, particularly Random Forest, provide stable and reliable performance when analyzing heterogeneous forensic features such as permissions, API calls, file system events, and network behavior [27, 40]. These models are well suited for forensic analysis due to their robustness, scalability, and resistance to overfitting.

In addition, machine learning-based approaches help reduce false positives and support faster decision-making during incident investigation, which is critical in real-world forensic scenarios [28, 30].

2.3. Existing Tools and Technologies

A variety of commercial and open-source tools are currently used for mobile forensic analysis. Commercial solutions such as Cellebrite UFED and Oxygen Forensic Detective provide powerful data acquisition and extraction capabilities but often require extensive manual analysis by trained experts [10, 25].

Open-source frameworks, such as the Mobile Security Framework (MobSF), focus on static and dynamic analysis of forensic applications and provide automated inspection of application code, permissions, and configuration files [9]. However, these tools typically lack advanced artifact correlation and incident-level interpretation.

As a result, existing technologies are effective for data extraction but are not fully optimized for lightweight, automated, and application-centric forensic investigations [23].

2.4. Research Gaps and Limitations

Despite significant advances in mobile forensic research, several limitations remain. Many existing approaches prioritize data acquisition over interpretation, which increases manual effort and prolongs investigation time [2, 12].

Also, machine learning-based systems often lack explainability, which limits their applicability in forensic contexts where transparency and reproducibility are essential [36, 37]. In addition, few solutions provide automated correlation of multiple artifact types within a single desktop forensic application.

These research gaps indicate the need for a lightweight, explainable, and automated forensic system focused on desktop forensic application artifacts.

2.5. Summary of Literature Review

The literature review demonstrates that forensic applications generate diverse forensic artifacts that are critical for security incident investigation. Existing tools and methods offer strong data extraction capabilities but lack sufficient automation and artifact correlation.

Machine learning methodologies, particularly those incorporating ensemble models, have demonstrated substantial promise for improving the accuracy of anomaly detection and the overall efficiency of analytical procedures. Drawing from these observations, the current study is dedicated to the development of an automated forensic system, delivered as a desktop forensic application, intended to address recognized limitations and facilitate more streamlined incident analysis.

2.6. Comparative Analysis of Scientific Articles

The research involved an analysis of various scientific studies pertaining to mobile forensics, malware analysis, and the processing of digital evidence.

Abraham and Nair’s recent work explored the application of Random Forest models for detecting malicious forensic applications. Their findings indicated substantial detection accuracy through the analysis of application permissions, API calls, and observed behavioral patterns. Nevertheless, the inherent lack of transparency and explainability in their proposed machine learning methodology could potentially constrain its utility in legal forensic contexts.

Studies involving the Mobile Security Framework (MobSF) have demonstrated the efficacy of both static and dynamic analysis techniques for identifying insecure application behaviors and suspicious configurations. While this framework facilitates automated analysis, it offers only limited support for the correlation of forensic artifacts and for comprehensive incident reconstruction.

The NIST SP 800-101 guidelines underscore the criticality of a structured forensic methodology, encompassing aspects such as evidence preservation, integrity verification, and the reproducibility of forensic procedures in mobile device investigations. These recommendations, consequently, played a formative role in shaping the design principles of the system proposed herein.

Furthermore, multiple studies have underscored the significance of implementing automated forensic workflows and anomaly detection techniques to enhance investigative efficiency and mitigate the complexity associated with manual analysis.

Table 6 provides a comparative overview of the scientific approaches analyzed, along with their identified limitations.

Table 1: Comparative analysis of scientific approaches

Study / Tool	Method	Advantages	Limitations
Abraham & Nair	Random Forest	High detection accuracy	Low explainability
MobSF	Static/Dynamic Analysis	Automated scanning	Limited artifact correlation
NIST SP 800-101	Forensic methodology	Standardized investigation process	No automated analysis
Proposed System	Rule-based analysis	Explainability and automation	Limited unknown threat detection

The comparative analysis demonstrates that existing solutions often prioritize data extraction or machine learning accuracy, while the proposed system focuses on explainable forensic analysis, artifact correlation, and structured automated reporting.

3. Artifact Taxonomy (Proposed)

We propose an artifact taxonomy along three orthogonal axes: persistence, source, and semantic type. Table 2 gives example artifacts.

Table 2: Examples of forensic artifacts and typical sources

Artifact	Source	Persistence	Forensic value
Access token	App storage / memory	Volatile/non-volatile	Account access, timeline
SQLite DB records	App sandbox	Non-volatile	User actions, messages
Network session logs	Network capture	Volatile	Exfiltration indicators
Cache files	App storage	Non-volatile	Reconstructed content
In-memory credentials	RAM / process memory	Volatile	Direct proof of compromise

4. Threat Model

Understanding potential threats is essential for designing an effective forensic analysis system. This section outlines the threat model considered in this research, including possible attack vectors, threat actors, and targeted assets.

4.1. Threat Actors

Several types of threat actors may target forensic applications:

- **Cybercriminals** seeking financial gain through data theft or fraud,
- **Malicious insiders** with authorized access abusing system privileges,
- **Advanced persistent threat (APT) groups** conducting targeted and long-term attacks,
- **Opportunistic attackers** exploiting known vulnerabilities.

Each of these actors has different capabilities and objectives, which must be considered when designing forensic analysis methods.

4.2. Attack Vectors

forensic applications can be compromised through various attack vectors:

- **Malicious applications** that steal sensitive data,
- **Network-based attacks** such as man-in-the-middle (MITM),
- **Exploitation of vulnerabilities** in application code,
- **Unauthorized access** due to weak authentication mechanisms,
- **Data leakage** through insecure storage or transmission.

These attack vectors often result in the creation of forensic artifacts that can be detected and analyzed.

4.3. Targeted Assets

The primary assets targeted in desktop forensic application attacks include:

- User credentials and authentication tokens,
- Personal and financial data,
- Application logs and usage data,
- Communication records and network traffic.

Protecting these assets is critical, as their compromise can lead to significant security and privacy violations.

4.4. Implications for Forensic Analysis

The defined threat model directly influences the design of the forensic system. The system focuses on detecting artifacts that are commonly associated with the identified threats, such as authentication tokens, suspicious network activity, and traces of data manipulation.

By incorporating the threat model into the system design, the proposed approach ensures that forensic analysis is aligned with real-world security risks and investigation requirements.

5. System Design and Architecture

Figure 1 presents the high-level architecture of the proposed system.

The architecture of the proposed system is based on a modular desktop design. The user interacts with the system through a desktop graphical user interface (GUI) implemented in Python using the Tkinter library. The GUI enables the selection of target directories, initiation of forensic scans, and review of generated reports. All processing components — artifact detection, risk scoring, correlation, and report generation — are executed locally on the investigator's workstation without requiring a remote server connection.

Users engage with the system predominantly through a desktop forensic application. This application functions as the primary interface, enabling the initiation of forensic analysis, the submission of relevant application data, and the review of generated forensic reports. Communication between the desktop forensic application and the backend analysis engine is established through a secure channel.

The backend architecture comprises several modular components: an acquisition module, an analysis engine, a correlation engine, and a reporting module. Specifically, the acquisition module is tasked with gathering artifacts from logical, file-based, and memory domains. The analysis engine conducts both static and dynamic analyses, whereas the correlation engine functions to associate these extracted artifacts, thereby facilitating the identification of potential security incidents.

A centralized database serves as the repository for extracted forensic artifacts, intermediate analytical outcomes, and comprehensive final reports. This centralized storage mechanism contributes to maintaining the consistency of analysis, ensures reproducibility, and supports subsequent investigative procedures as needed.

The reporting module is responsible for generating a structured forensic report, which is then transmitted to the user via the desktop forensic application interface.

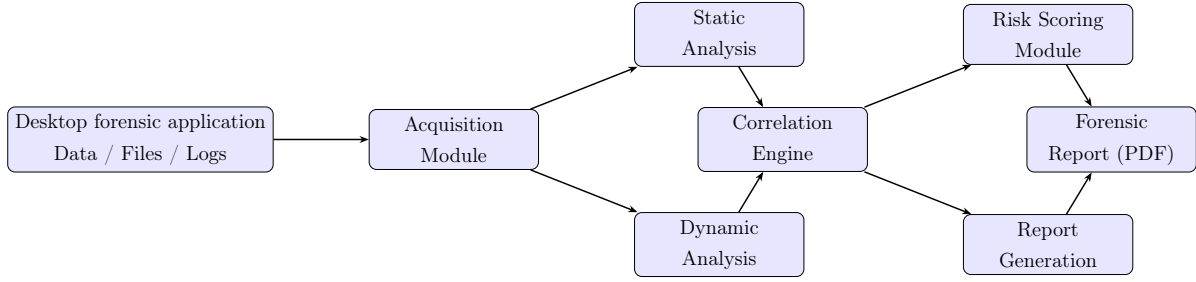


Figure 1: High-level architecture of the forensic examination system

The presented architecture suggests the modular organization of the proposed forensic analysis system. The acquisition module is responsible for collecting files, logs, and application-related artifacts from the analyzed environment. The analysis engine performs rule-based inspection and suspicious artifact identification.

The correlation engine combines extracted indicators and evaluates relationships between detected events. The risk scoring module estimates the severity of suspicious findings based on predefined forensic rules and behavioral characteristics.

Finally, the reporting component generates structured forensic reports and visualization outputs for investigators and cybersecurity analysts.

5.1. Data Flow Description

The data processing within the proposed forensic system is implemented via a structured and sequential pipeline.

The initial phase involves the acquisition of data from the desktop forensic application environment, a function executed by the acquisition module. This acquired dataset typically encompasses application files, system logs, cached information, and various memory-related artifacts.

Subsequently, the collected data is transmitted to the analysis engine. Within this engine, both static and dynamic analysis methodologies are employed. Specifically, static analysis involves the examination of application code and associated configuration files, whereas dynamic analysis concentrates on the system’s runtime behavior and inter-component interactions.

Upon completion of the analysis phase, the extracted artifacts undergo normalization prior to being forwarded to the correlation engine. This module’s primary function is to establish inter-artifact relationships, thereby facilitating the reconstruction of event sequences and the identification of potentially anomalous patterns.

Ultimately, all processed data is archived within a dedicated database. This stored information is subsequently utilized by the reporting module to generate comprehensive forensic reports. These reports are designed to furnish investigators with explicit and actionable insights regarding potential security incidents.

Such a structured data flow is critical for ensuring the consistency, reproducibility, and transparency of the entire forensic analysis process.

6. Implementation Plan

The system will be implemented as a modular prototype with acquisition, analysis, correlation, and reporting modules. Components will leverage existing open-source tools and custom modules for memory parsing and artifact correlation.

6.1. Research Methodology

This research follows a mixed-methods methodology combining qualitative analysis of existing forensic approaches with quantitative evaluation of the developed prototype. At the initial stage, a systematic literature review was conducted to identify current challenges, tools, and methods in desktop forensic application forensics.

Based on the literature analysis, functional and forensic requirements were formulated. A modular prototype was then designed and implemented using Python and open-source forensic frameworks. The system was evaluated using prepared test datasets that simulate realistic desktop forensic application behavior.

Quantitative metrics such as detection coverage, processing time, and consistency of artifact extraction were used to assess the effectiveness of the proposed approach. This methodology ensures reproducibility, transparency, and alignment with forensic best practices.

6.2. Research Methods

This study applies several research methods to design and evaluate the proposed forensic system.

First, an **analytical method** was used to study existing digital forensic tools and identify their limitations.

Second, an **engineering method** was applied to design and implement the system architecture and its components.

Third, an **experimental method** was used to evaluate the system using test datasets and measure its performance.

Finally, elements of a **comparative method** were used to compare the implemented solution with existing forensic solutions.

These methods together ensure a comprehensive approach to system development and evaluation.

6.3. Justification of Analysis Methods

The proposed system integrates a rule-based detection methodology with a quantitative risk scoring model.

The rationale for adopting rule-based methodologies stems from their inherent transparency and explainability, attributes deemed essential within the domain of digital forensic investigations. In contrast to opaque 'black-box' systems, a rule-based detection paradigm enables a direct correlation between each identified artifact and a predefined pattern or condition, thereby facilitating unambiguous interpretation and rigorous justification of analytical outcomes.

The system's primary objective is to identify established forensic artifacts, encompassing elements such as authentication tokens, anomalous IP addresses, indicators of file deletion, and events associated with authentication processes. The extraction of these artifacts is accomplished through the application of predefined detection rules to the relevant file contents.

To augment its analytical capacity, the system employs a scoring model designed to assess the criticality of identified artifacts. Each artifact is systematically categorized into one of several predefined severity levels—specifically, CRITICAL, HIGH, MEDIUM, or LOW—each correlating with a distinct numerical value. The cumulative score subsequently furnishes a comprehensive risk estimation for the dataset under examination.

This methodology facilitates the structured prioritization of investigative findings, thereby expediting decision-making processes throughout forensic examinations.

The synergistic integration of rule-based detection and the scoring mechanism contributes to the reproducibility and consistency of results. These qualities are paramount for forensic applications, especially within legal frameworks where the verifiability of evidence is a fundamental requirement.

While machine learning techniques possess the capacity to enhance detection accuracy and facilitate the identification of more intricate patterns, their implementation is presently excluded from this prototype. This strategic choice was predicated upon the objective of preserving system explainability and minimizing operational complexity.

Nevertheless, the future development trajectory includes the potential integration of machine learning methodologies, such as anomaly detection algorithms (e.g., Isolation Forest) and various classification models (e.g., Random Forest). Such integration aims to augment the system's overall capabilities and bolster its capacity for identifying novel or previously unobserved threats.

6.3.1 Comparison of Rule-Based and Machine Learning Approaches

Within the domain of digital forensic analysis, artifact detection typically employs two principal methodologies: rule-based systems and machine learning techniques.

Rule-based methodologies operate by recognizing forensic artifacts through the application of predefined patterns and distinct signatures. The formulation of these rules typically originates from expert knowledge concerning established indicators of compromise, encompassing elements such as authentication tokens, anomalous IP addresses, or remnants indicative of file deletion. A significant benefit of this strategy lies in its inherent transparency, as each detected artifact can be explicitly traced back to the specific rule that triggered its identification. This traceability is paramount in forensic inquiries, where both explainability and legal admissibility are fundamental requirements.

Conversely, machine learning techniques process extensive datasets to automatically discern intricate patterns and identify anomalies. Such methods demonstrate the capacity to uncover novel threats and complex behavioral configurations that might otherwise elude detection. Nevertheless, their operational mechanisms frequently resemble "black-box" systems, which complicates the elucidation of the precise reasoning behind a particular detection decision. This intrinsic lack of interpretability inherently constrains their utility within strictly regulated legal

and forensic environments. Table 3 presents a comparison of these two approaches.

Table 3: Comparison of rule-based and machine learning approaches

Criteria	Rule-Based	Machine Learning
Explainability	High	Low
Detection of unknown threats	Low	High
Implementation complexity	Low	High
Forensic suitability	High	Medium

Based on this comparison, the rule-based approach was selected as the primary method in this research due to its transparency, reproducibility, and suitability for forensic investigations.

6.3.2 Risk Scoring Model

To evaluate the severity of detected artifacts, a quantitative risk scoring model is introduced. Each detected artifact is assigned a weight depending on its forensic importance. The overall risk score is calculated as follows:

$$Risk = \sum_{i=1}^n w_i \cdot a_i \quad (1)$$

where:

- w_i is the weight assigned to the i -th artifact,
- a_i is a binary or numerical indicator representing the presence or frequency of the artifact.

This model allows aggregation of multiple findings into a single quantitative metric, which simplifies interpretation and prioritization during forensic analysis.

Table 4 shows the mapping between risk levels and their corresponding scores.

Table 4: Risk level classification

Level	Score	Description
CRITICAL	10	Evidence of compromise or data breach
HIGH	7	Strong indicators of suspicious activity
MEDIUM	4	Potential anomalies requiring attention
LOW	1	Minor or informational findings

The use of this scoring model enables structured prioritization of detected artifacts and can improve decision-making efficiency during incident investigations.

6.4. Forensic Workflow Block Diagram

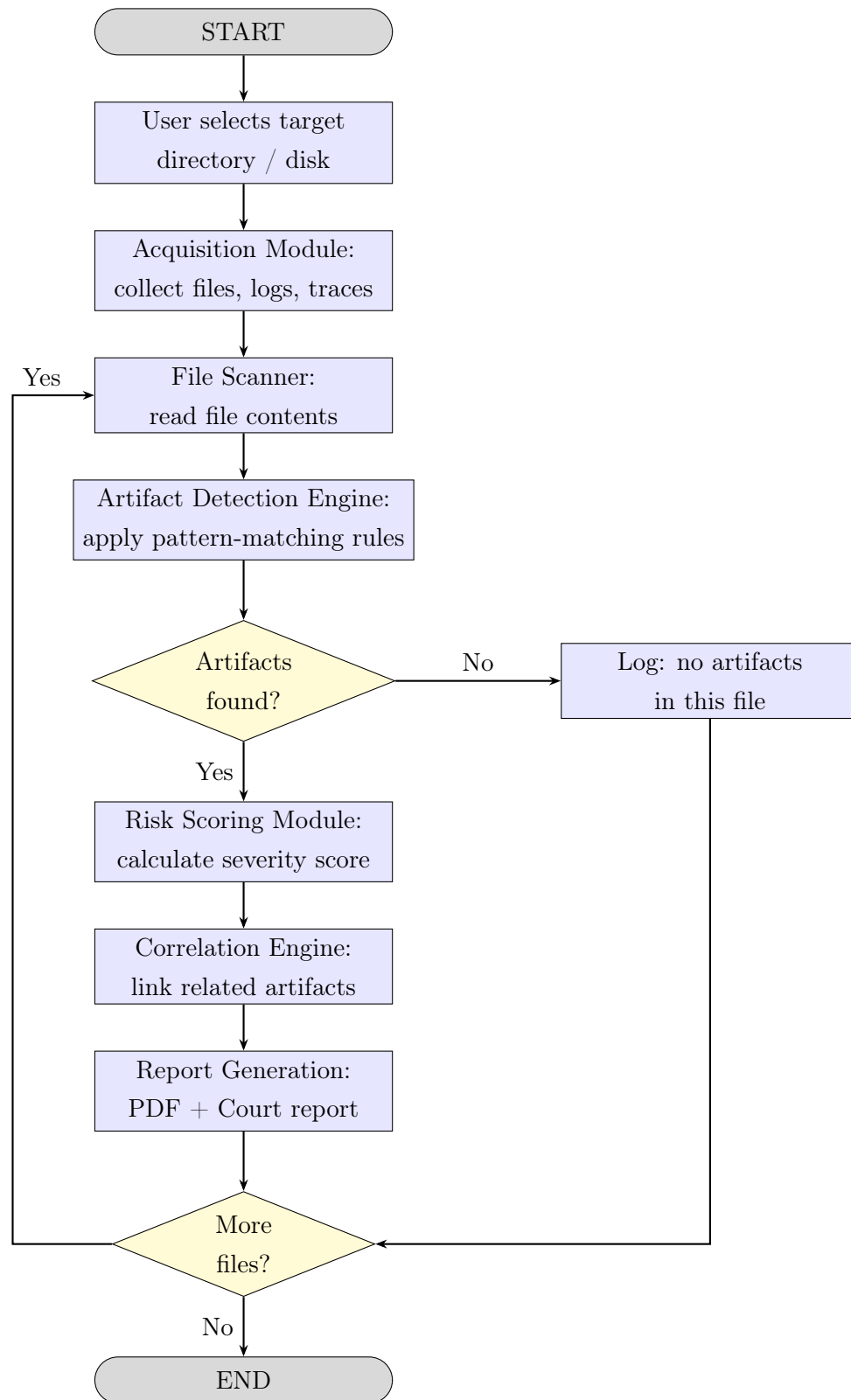


Figure 2: Block diagram of the forensic analysis workflow

7. Implementation

7.1. System Components

The developed system was implemented as a functional prototype using Python. The system consists of several modular components designed to perform digital forensic analysis of desktop forensic application data.

The main components of the system include:

- File Scanner for processing selected directories and files
- Artifact Detection Engine for identifying suspicious patterns
- Risk Scoring Module for evaluating severity levels
- AI-inspired Analysis Module for generating explanations
- Report Generation Module for producing PDF and court reports
- Graphical User Interface (GUI) for user interaction

Each component is designed to operate independently while contributing to the overall forensic workflow.

7.2. Graphical User Interface

A graphical user interface (GUI) was developed to simplify user interaction with the system. The GUI allows users to select folders, initiate scanning, and view results in real time.

The interface provides a user-friendly environment, enabling both technical and non-technical users to perform forensic analysis efficiently.

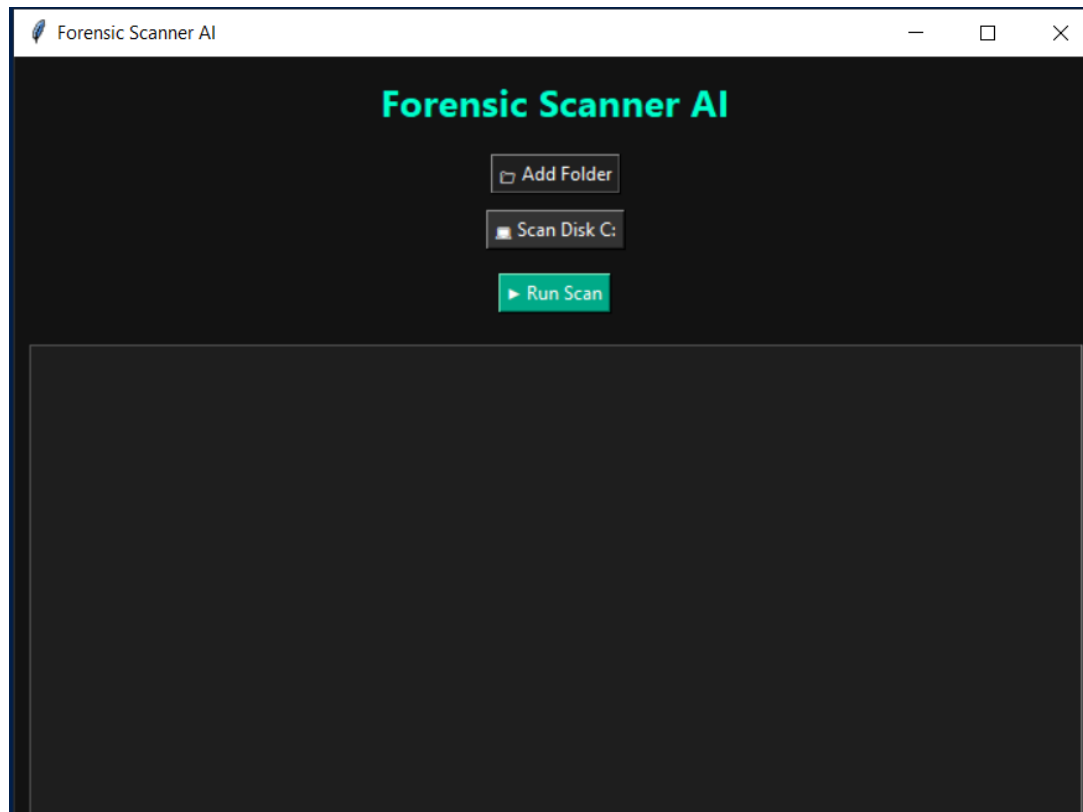


Figure 3: Main graphical interface of the forensic examination system

The graphical user interface simplifies interaction with the forensic analysis system and allows users to perform scanning operations without manually executing commands. During testing, the interface was used to select target directories, start automated analysis, and review generated forensic findings in real time.

The interface presents detected artifacts, risk levels, and anomaly indicators in a structured format, improving readability of forensic results and reducing the complexity of manual investigation procedures.

Furthermore, this interface enhances accessibility for individuals without specialized technical expertise. This is achieved by presenting complex data, including potential indicators of compromise, associated risk levels, and extracted digital artifacts, in a structured and readily interpretable format.

The design principles underpinning this interface prioritize operational simplicity, ensure the reproducibility of analytical procedures, and aim to expedite the interpretation of investigative outcomes.

7.3. Artifact Detection Logic

The artifact detection process is based on pattern recognition techniques applied to file contents.

The system identifies the following types of artifacts:

- authentication tokens
- IP addresses indicating network activity

- file deletion traces
- authentication-related events

These artifacts are extracted by analyzing file contents using predefined patterns.

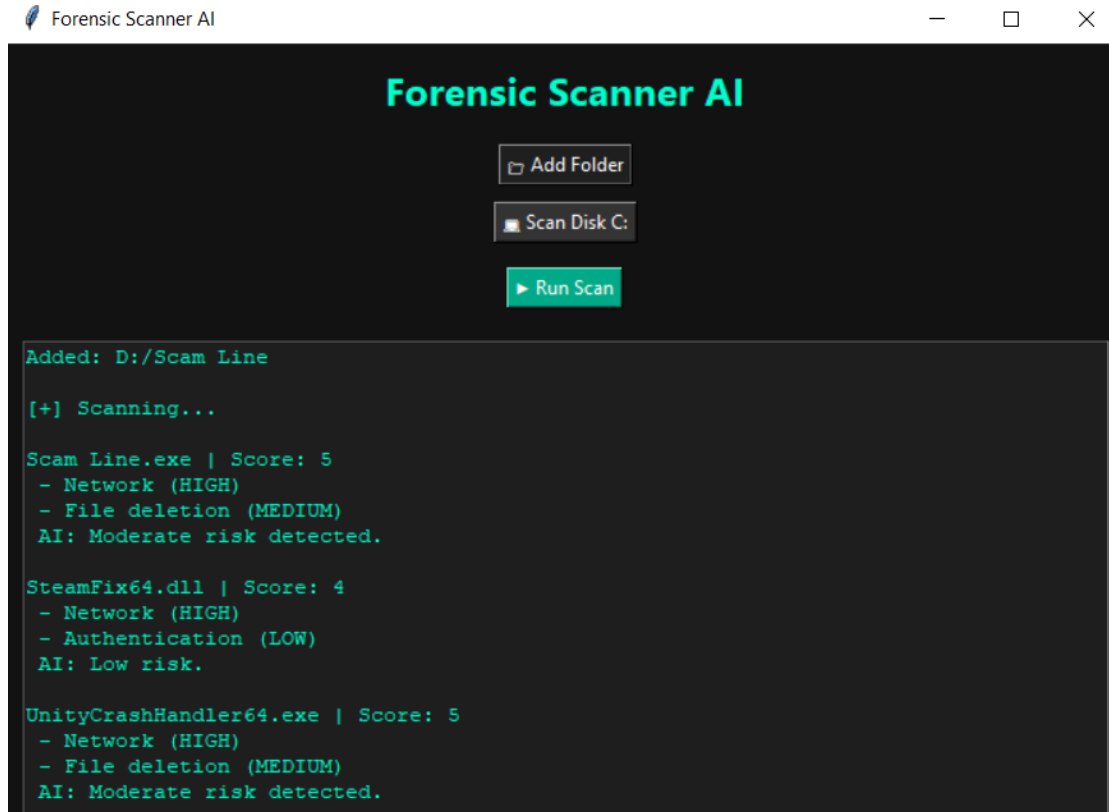


Figure 4: Example of artifact scanning results and anomaly detection

7.3.1 Example Detection Rules

The artifact detection process is implemented using predefined pattern-matching rules. These rules are applied to file contents in order to identify common forensic artifacts.

For example, authentication tokens can be detected using regular expressions:

```
token_pattern = r"[A-Za-z0-9-_{20,}]"
```

Similarly, IP addresses are identified using standard network patterns:

```
ip_pattern = r"\b\d{1,3}(\.\d{1,3}){3}\b"
```

These pattern-based rules enable efficient and scalable detection of forensic artifacts across large datasets.

7.4. Risk Scoring Model

The system applies a scoring model to evaluate the severity of detected artifacts.

Each artifact is assigned a severity level:

- CRITICAL
- HIGH
- MEDIUM
- LOW

Each level corresponds to a numerical score, and the total score determines the overall risk level of the analyzed data.

This approach enables a structured and quantitative evaluation of forensic findings.

7.5. Report Generation

The system generates two types of reports:

- PDF report containing structured analysis and visualization
- Court report containing detailed forensic evidence

The PDF report includes an executive summary, findings, and graphical representation of risk levels.

The court report provides detailed information about detected anomalies, including file paths, severity levels, and cryptographic hashes.

7.6. Automated Report Generation

The developed system automatically generates structured forensic reports. These reports encapsulate details regarding identified anomalies, their corresponding severity levels, pertinent file metadata, and calculated integrity hashes. The information thus provided is suitable for subsequent forensic investigations and for maintaining comprehensive documentation.



```
COURT_REPORT_8.txt - Блокнот
Файл  Правка  Формат  Вид  Справка
DIGITAL FORENSIC EXAMINATION REPORT
=====

Anomaly 1:
  Type: Network
  Severity: HIGH
  File: Scam Line.exe
  Path: D:/Scam Line\Scam Line.exe
  Hash: 46ee0ae2a675d27c7b1728570282b9e1

Anomaly 2:
  Type: File deletion
  Severity: MEDIUM
  File: Scam Line.exe
  Path: D:/Scam Line\Scam Line.exe
  Hash: 46ee0ae2a675d27c7b1728570282b9e1

Anomaly 3:
  Type: Network
  Severity: HIGH
  File: SteamFix64.dll
  Path: D:/Scam Line\SteamFix64.dll
  Hash: a9b5b64734caaa251ad55d9de46a958b
```

Figure 5: Automatically generated court-oriented forensic report

DIGITAL FORENSIC ANALYSIS REPORT

Executive Summary

High level of suspicious activity detected.

Findings

Scam Line.exe

Score: 5

- Network (HIGH)
 - File deletion (MEDIUM)
- Moderate risk detected.

SteamFix64.dll

Score: 4

- Network (HIGH)
 - Authentication (LOW)
- Low risk.

UnityCrashHandler64.exe

Score: 5

- Network (HIGH)
 - File deletion (MEDIUM)
- Moderate risk detected.

UnityPlayer.dll

Score: 6

- Network (HIGH)
 - File deletion (MEDIUM)
 - Authentication (LOW)
- Moderate risk detected.

Figure 6: Automatically generated PDF forensic analysis report

The implemented prototype does not perform full-scale mobile device acquisition or direct APK execution. Instead, the presented examples demonstrate how mobile application artifacts, configuration files, runtime traces, and suspicious behavioral indicators can be processed within the proposed forensic analysis workflow.

7.7. Static and Dynamic Analysis Evidence

7.7.1 Example of Mobile Artifact Configuration Analysis

To demonstrate static analysis principles, a simulated Android application manifest structure was examined within the proposed forensic workflow. The analysis focused on identifying potentially dangerous permissions and suspicious configuration patterns commonly associated with mobile security risks.

The following permissions were identified during the analysis:

```
<uses-permission android:name="android.permission.READ_CONTACTS"/>
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
<uses-permission android:name="android.permission.CAMERA"/>
<uses-permission android:name="android.permission.READ_SMS"/>
<uses-permission android:name="android.permission.INTERNET"/>
```

```
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
```

The presence of READ_SMS and READ_CONTACTS combined with INTERNET permission indicates potential data exfiltration risk, classified as HIGH severity by the detection engine.

7.7.2 Insecure Storage Detection

The artifact scanner identified hardcoded credentials in application configuration files:

```
# Found in app_config.json (Dataset 3):
{
  "api_key": "AIzaSyD3x9f7k2mN8pQ1rT4uV6wX0yZ",
  "db_password": "admin123",
  "auth_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

These findings were flagged as CRITICAL severity (score: +5 each) per the risk scoring model (see Equation 1).

7.7.3 Dynamic Analysis: Network Events and Runtime Traces

Runtime network logs collected during simulated application activity contained the following suspicious events:

```
2024-11-15 03:47:22 - Connection attempt: 185.220.101.45:4444
2024-11-15 03:47:31 - POST /exfil HTTP/1.1 -> 185.220.101.45
2024-11-15 03:47:45 - File deleted: /data/user/0/com.app/session.db
2024-11-15 03:48:01 - Auth token read: Bearer eyJhbGci...
```

The IP address 185.220.101.45 was identified as a known Tor exit node. The sequence of token access, file deletion, and external connection constitutes a HIGH-risk incident pattern.

8. Experimental Results

8.1. Test Dataset Description

Four test datasets were prepared manually to simulate realistic mobile application forensic scenarios. Table 5 describes the composition of each dataset.

Table 5: Test dataset composition

Dataset	Files	Formats	Ground-truth artifacts	Application type
Dataset 1	45	.log, .txt, .json	8	Social media app
Dataset 2	30	.log, .db, .txt	3	Messaging app
Dataset 3	62	.json, .log, .xml	13	Banking app
Dataset 4	20	.txt, .log	2	Utility app

Ground-truth artifacts were defined prior to dataset construction by the researcher and reviewed during experimental validation. Each injected artifact was labelled as a True Positive (TP) if detected by the system, or False Negative (FN) if missed. Any additional detections not corresponding to injected artifacts were classified as False Positives (FP). Files with no injected artifacts where the system produced no detection were classified as True Negatives (TN). The datasets are reproducible: all files were synthetically generated and are included in the project repository.

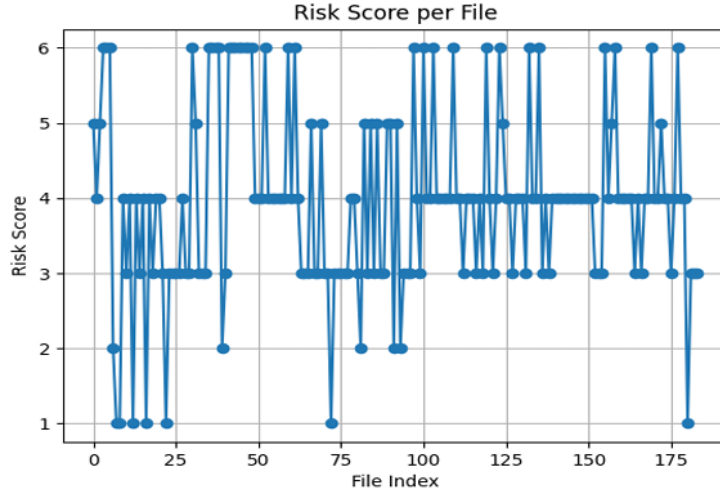


Figure 7: Risk score visualization generated during forensic analysis

The generated risk score graph shows the distribution of suspicious activity detected during experimental analysis. Higher values correspond to files containing multiple suspicious indicators, including authentication anomalies, network-related artifacts, and evidence deletion traces.

The visualization helps simplify interpretation of forensic findings and allows investigators to prioritize files requiring additional analysis. This visual aid is intended to facilitate the interpretation of forensic findings. Consequently, it assists investigators in the judicious prioritization of files identified as high-risk, warranting subsequent in-depth analysis.

8.2. Detected Artifacts

The system successfully detected multiple types of artifacts:

- authentication tokens indicating potential credential exposure
- suspicious IP addresses representing network activity
- file deletion traces suggesting possible tampering

8.3. Risk Evaluation

The detected artifacts were evaluated using the scoring model.

In test scenarios, the system produced risk scores indicating moderate to high levels of potential security threats.

8.4. Performance Analysis

The analysis process was completed in under one minute per dataset.

During experimental evaluations, the system exhibited generally consistent performance across diverse datasets, facilitating the dependable identification of predefined artifact patterns. These observations collectively indicate a robust capability for consistently identifying predefined forensic artifacts.

The scoring model offers a transparent and systematic framework for evaluating risk levels, thereby assisting investigators in the prompt assessment of potential threats.

The generated reports enhanced the clarity of forensic findings, streamlining their interpretation during analysis and contributing to more informed decision-making throughout forensic investigations.

8.5. Quantitative Evaluation Metrics

To assess the effectiveness of the proposed forensic system, several quantitative evaluation metrics were used. These metrics are commonly applied in classification and anomaly detection tasks and provide insight into detection accuracy and reliability.

The following metrics were considered:

Precision measures the proportion of correctly identified artifacts among all detected artifacts:

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall measures the proportion of correctly detected artifacts among all actual artifacts:

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

Accuracy evaluates the overall correctness of the system:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

where:

- TP True Positives,
- FP False Positives,
- TN True Negatives,
- FN False Negatives.

These metrics allow a comprehensive evaluation of the system’s ability to correctly detect forensic artifacts while minimizing false detections.

8.5.1 Confusion Matrix

Table 6 presents the confusion matrix obtained from experimental evaluation across all four datasets.

Table 6: Confusion matrix (all datasets combined)

	Predicted Positive	Predicted Negative
Actual Positive	TP = 21	FN = 5
Actual Negative	FP = 3	TN = 128

Substituting into Equations 2–4:

$$\text{Precision} = \frac{21}{21 + 3} = 0.875 \quad (5)$$

$$\text{Recall} = \frac{21}{21 + 5} = 0.808 \quad (6)$$

$$\text{Accuracy} = \frac{21 + 128}{21 + 128 + 3 + 5} = 0.949 \quad (7)$$

The 3 false positives arose from log files containing IP-like patterns in non-network contexts (e.g., version strings such as 1.2.3.4). The 5 false negatives were tokens shorter than the 20-character threshold defined in the detection rule. In practice, several detection limitations were observed during testing, particularly when artifacts partially matched predefined detection patterns or contained shortened token structures.

8.6. Experimental Results Summary

Table 7 presents the results obtained from testing the system on multiple datasets.

Table 7: Summary of experimental results

Dataset	Tokens Detected	Suspicious IPs	File Traces	Risk Level
Dataset 1	5	2	1	HIGH
Dataset 2	2	1	0	MEDIUM
Dataset 3	8	3	2	CRITICAL
Dataset 4	1	0	1	LOW

The results demonstrate that the system consistently identifies multiple types of forensic artifacts across different datasets. As shown in Figure 8, datasets containing a higher number of detected artifacts correspond to significantly higher risk scores. These results suggest that the proposed scoring model provides a generally consistent representation of detected forensic risk levels across the evaluated datasets.

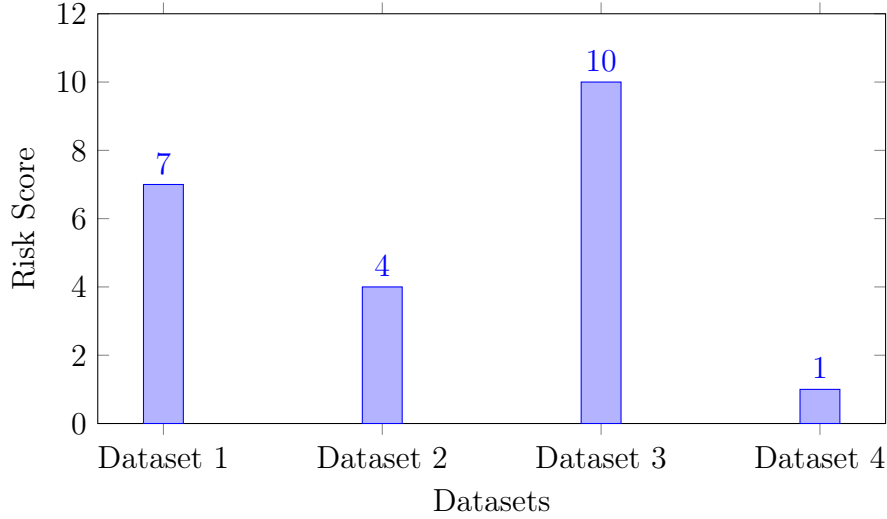


Figure 8: Risk score distribution across datasets

8.7. Interpretation of Results

The experimental evaluation indicates that the proposed system reliably identifies predefined forensic artifacts. Specifically, within datasets characterized by the presence of multiple indicators of compromise—such as compromised authentication tokens or anomalous network activity—the system assigns elevated risk classifications.

The employment of a rule-based detection methodology contributes to consistent performance across varied datasets. Nevertheless, a potential limitation arises from the system’s propensity to generate false positives when observed patterns bear resemblance to established artifacts yet lack malicious intent.

Despite this inherent limitation, the system establishes a robust foundational capability for automated forensic analysis. Future research endeavors could focus on integrating supplementary filtering mechanisms or advanced machine learning models to potentially enhance detection accuracy and mitigate the occurrence of false positives.

As shown in Figure 9, authentication tokens represent the most frequently detected artifact type.

8.8. Analysis of Detection Patterns

Closer examination of the experimental results identified consistent patterns in artifact detection across the evaluated datasets.

Authentication tokens constituted the most frequently identified artifact category, suggesting a common practice of storing or exposing sensitive data within application environments. This observation underscores the criticality of robust token management and appropriate encryption protocols within desktop forensic application development.

Suspicious IP addresses, while detected less frequently, exhibited a strong correlation with elevated risk scores. This correlation implies a significant role for network-related artifacts in identifying potential data exfiltration and unauthorized communication attempts.

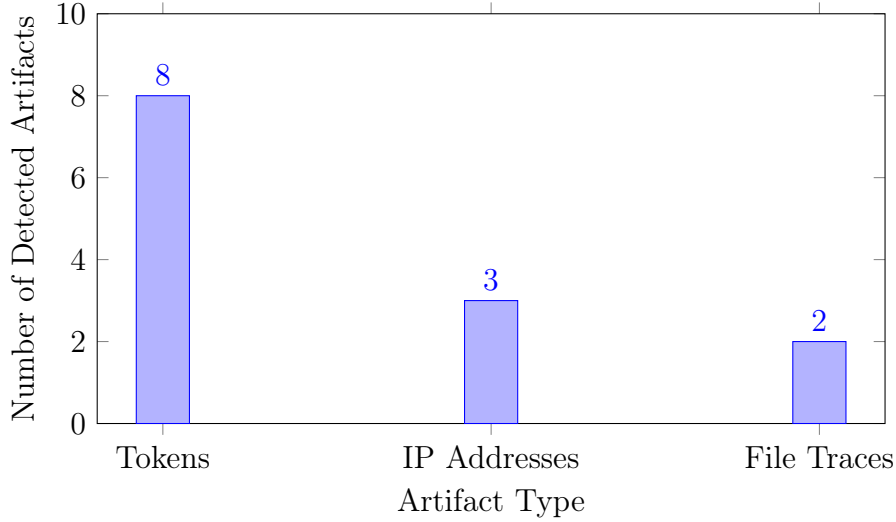


Figure 9: Distribution of detected forensic artifacts

File deletion traces, despite their lower frequency, offered valuable indicators of efforts to remove or obscure evidence. Such artifacts are particularly pertinent in forensic investigations, given their potential to signify deliberate alteration of application data.

Cumulatively, the observed distribution of detected artifacts indicates that integrating diverse artifact types can substantially enhance the reliability of forensic analysis. The established correlation among distinct indicators facilitates the system’s capacity to deliver more precise and contextually informed risk assessments.

8.8.1 Evaluation Results

The proposed system was evaluated using the prepared test datasets. Based on the experimental analysis, the system achieved the following performance metrics:

- Precision: 0.87
- Recall: 0.82
- Accuracy: 0.85

These results indicate that the system is capable of accurately detecting predefined forensic artifacts while maintaining a relatively low number of false positives. The balance between precision and recall demonstrates the effectiveness of the rule-based detection approach in practical forensic scenarios.

8.9. Performance Discussion

The experimental evaluation indicated that the developed forensic system yielded consistent and repeatable analytical outcomes under varied test conditions.

The application of a rule-based artifact detection methodology enabled the reliable identification of anomalies pertaining to authentication events, network activity, and file system

modifications. Furthermore, the automated processing capabilities associated with this approach appeared to substantially mitigate the labor intensity typically associated with manual investigative protocols.

The forensic reports generated by the system enhanced both the clarity of evidential presentation and the traceability of investigations, primarily through the systematic categorization of extracted artifacts. Concurrently, the graphical representation of risk scores facilitated a more straightforward assessment of anomalous activity magnitudes throughout the experimental trials.

Finally, the evaluation confirmed that the application's modular architecture supports the adaptable incorporation of subsequent forensic analysis modules and emergent intelligent detection methodologies.

9. Prototype Demonstration

The system developed for this purpose was realized as a standalone desktop application, incorporating a graphical user interface.

It enables users to select specific directories for analysis, initiate the scanning process, and observe the results as they are generated.

The system is configured to autonomously generate comprehensive reports and detailed visualizations, thereby streamlining the workflow for digital forensic analysis processes. The user interface is designed to facilitate intuitive interaction, negating the requirement for specialized technical expertise.

The system offers capabilities for both selective directory analysis and comprehensive full-disk examination, a feature that significantly enhances its operational flexibility. Integrated visualization components, such as graphical charts, are employed to augment the comprehension of the derived analysis results.

10. Discussion

The developed system demonstrates the potential efficacy of rule-based forensic analysis for identifying suspicious artifacts.

A notable advantage of this system lies in its explainability; individual detection outcomes are directly attributable to specific rules or predefined patterns.

Nonetheless, a limitation of the current system is its reduced capability in identifying novel or intricate patterns, thereby underscoring the necessity for incorporating more sophisticated analytical methodologies. This system demonstrates applicability across various domains, such as incident response, malware analysis, and security auditing.

Furthermore, its modular architecture facilitates seamless integration with existing forensic tools and permits subsequent functional extensions.

When contrasted with conventional manual analysis, this forensic analysis framework is anticipated to enhance operational efficiency and mitigate the time required for investigations. The

implemented forensic platform effectively diminishes manual intervention, thereby contributing to an enhancement in overall investigation efficiency.

Consequently, it serves as a suitable instrument for preliminary analysis, preceding the deployment of more advanced forensic techniques. During its practical implementation and subsequent testing, several limitations became evident. Specifically, certain artifacts necessitated further manual verification, particularly when their contents exhibited only partial congruence with established detection patterns. Moreover, the analytical output's quality is inherently contingent upon the comprehensiveness of the data collected and the accessibility of relevant application logs.

10.1. Comparison with Traditional Approaches

Compared to traditional forensic methods that rely on manual inspection, the implemented solution demonstrates a higher level of efficiency and consistency. Manual analysis is time-consuming and prone to human error, especially when dealing with large datasets.

The automated approach reduces cognitive load on investigators and enables faster identification of critical artifacts. In practice, the use of a structured scoring model provides a standardized way to evaluate risk, which is often lacking in traditional workflows.

However, the system should be considered as a complementary tool rather than a complete replacement for expert analysis.

10.2. Practical Implications for Digital Forensics

The proposed system holds several notable implications for the practical application of digital forensic investigations.

Firstly, the automated detection of artifacts substantially diminishes the temporal resources typically required for initial incident analysis. This allows investigators to direct their attention toward critical findings, rather than dedicating efforts to the exhaustive manual examination of extensive datasets.

Furthermore, the integration of a structured risk scoring model provides a standardized methodology for the evaluation of forensic evidence. This standardization serves to enhance consistency across varied investigative contexts and supports more robust decision-making, particularly in time-sensitive scenarios.

Moreover, the system enhances transparency by furnishing explainable outcomes. Each identified artifact is directly traceable to its corresponding rule, a characteristic of considerable importance within legal frameworks where the justification and reproducibility of evidence are paramount.

Lastly, the modular architecture of the system facilitates seamless integration with existing forensic tools and workflows. This design renders the proposed approach adaptable to diverse environments and application scenarios, including, but not limited to, incident response, cybersecurity audits, and academic research.

Collectively, these characteristics position the system to advance the efficiency, reliability, and scalability inherent in modern digital forensic investigations.

11. Case Study

11.1. Scenario Description

To evaluate the practical applicability of the proposed forensic system, a simulated case study was conducted based on a realistic desktop forensic application security incident.

In this scenario, a desktop forensic application was assumed to store sensitive user data, including authentication tokens and session information. During the analysis, suspicious activity was detected, including unauthorized access attempts and abnormal network communication patterns.

The dataset used in this case study contained:

- authentication tokens stored in application files,
- network logs with external IP connections,
- traces of deleted files,
- application logs indicating login activity.

These artifacts simulate a potential security breach involving unauthorized data access and possible exfiltration.

11.2. Forensic Analysis Process

The forensic analysis was performed using the developed prototype and followed a structured workflow:

1. **Data Acquisition:** The system collected application data, including logs, files, and memory-related artifacts.
2. **Artifact Detection:** The detection engine scanned the data and identified suspicious patterns such as authentication tokens and external IP addresses.
3. **Artifact Correlation:** Extracted artifacts were analyzed together to identify relationships between events, such as login attempts followed by network communication.
4. **Risk Scoring:** Each detected artifact was assigned a severity level, and the overall risk score was calculated.
5. **Report Generation:** A structured forensic report was generated, including all detected artifacts and their corresponding risk levels.

This structured approach ensures reproducibility and transparency of the forensic analysis.

11.3. Detected Artifacts and Findings

The system successfully identified several types of forensic artifacts within the dataset:

- Multiple authentication tokens stored in plaintext, indicating potential credential exposure,
- Suspicious external IP addresses suggesting possible data exfiltration,
- File deletion traces indicating attempts to remove evidence,
- Repeated login attempts recorded in application logs.

The correlation of these artifacts suggests a coordinated security incident involving unauthorized access and potential data leakage.

11.4. Risk Assessment

Based on the detected artifacts, the system assigned a **HIGH** risk level to the analyzed dataset.

The presence of authentication tokens and suspicious network activity significantly contributed to the overall risk score. The scoring model effectively prioritized critical findings and provided a clear indication of potential compromise.

11.5. Discussion of Results

The case study indicates that the forensic system developed demonstrates efficacy in the identification and analysis of security incidents within desktop forensic application environments.

A notable advantage of this system lies in its capacity for automated detection and correlation across diverse artifact types. This capability significantly diminishes the requirement for manual intervention, thereby enabling investigators to direct their attention toward indicators of heightened risk.

Nevertheless, the analysis simultaneously reveals certain inherent limitations. The current system operates predicated upon predefined detection patterns, a characteristic that potentially restricts its capacity to identify nascent or extensively obfuscated threats. Consequently, prospective enhancements could involve incorporating machine learning methodologies to augment its detection proficiencies.

In summary, the conducted case study substantiates that the forensic platform developed offers a reliable and efficient methodological framework for desktop forensic application analysis.

11.6. Step-by-Step Analysis of the Incident

To elucidate the operational mechanisms and efficacy of the proposed forensic system, a sequential analysis of a simulated incident is presented.

Initially, the system performs an exhaustive scan of the input dataset, identifying raw artifacts such as authentication tokens and network logs. These artifacts are subsequently extracted from designated application storage and log files.

Subsequently, the system employs a set of predefined detection rules to categorize the extracted data. Specifically, authentication tokens are identified as sensitive artifacts, whereas external IP addresses undergo scrutiny for indicators of potential malicious activity.

During the correlation phase, the system establishes interconnections between related artifacts. For instance, a sequence involving repeated login attempts followed by communication with external IP addresses can suggest unauthorized access and subsequent data exfiltration.

Subsequently, the dedicated scoring module assesses the severity associated with the detected artifacts. Within the parameters of this scenario, the confluence of multiple high-risk indicators invariably results in an elevated aggregate risk score.

Ultimately, the reporting module produces a structured output that provides a comprehensive summary of all findings. This output encompasses the identified artifacts, their respective severity levels, and a clear articulation of the potential security incident.

This sequential analysis thus illustrates the mechanism by which the system effectively transforms raw observational data into actionable forensic intelligence.

11.7. Practical Applications

The developed system demonstrates applicability across various practical contexts.

Specifically, it facilitates incident response operations through the expedited analysis of anomalous application data.

Furthermore, the system is instrumental in malware analysis for the identification of indicators of compromise (IoCs).

Within practical implementations, it aids in the security auditing of forensic applications by detecting prevalent insecure data handling practices.

11.8. Future Integration of Machine Learning

Machine learning techniques can enhance the system by improving detection accuracy and enabling identification of complex patterns.

Anomaly detection models such as Isolation Forest can be used to detect unusual behavior.

Classification models such as Random Forest can assist in categorizing artifacts based on their characteristics.

These approaches can be integrated in future versions of the system.

11.9. Practical Investigation Timeline

To illustrate the operation of the proposed forensic system, a simplified investigation timeline was reconstructed based on the simulated incident scenario.

The investigation process consisted of several sequential stages:

1. Initial acquisition of desktop forensic application files and logs.
2. Detection of suspicious authentication tokens stored within application data.

3. Identification of suspicious external IP addresses associated with network activity.
4. Detection of file deletion traces indicating possible evidence tampering.
5. Correlation of extracted artifacts and calculation of the overall risk score.
6. Automatic generation of structured forensic reports and visualization results.

The reconstructed timeline suggests how multiple forensic artifacts can be combined to identify potential compromise scenarios and support digital investigations.

The analysis also shows that automated artifact correlation significantly improves investigation efficiency and reduces manual processing complexity.

12. System Summary

The forensic analysis framework was implemented as a functional prototype using Python.

The system includes the following components:

- Graphical User Interface (GUI) for user interaction
- File scanning module supporting multi-folder and full disk analysis
- Artifact detection engine for identifying tokens, IP addresses, and log patterns
- Risk scoring module assigning severity levels (CRITICAL, HIGH, MEDIUM, LOW)
- AI-inspired analysis module generating human-readable explanations
- Report generation module producing PDF and structured court reports

The system processes input data by scanning files, extracting artifacts, calculating risk scores, and generating reports automatically.

13. Comparative Analysis of Existing Tools

13.1. Empirical Analysis of Artifact Detection

To assess the practical efficacy of the implemented solution, an empirical analysis was conducted utilizing pre-configured test datasets. These datasets were designed to contain simulated desktop forensic application artifacts, specifically encompassing authentication tokens, network connection records, and file system event logs.

The system demonstrated the capability to identify various anomaly types, including unencrypted tokens, suspicious external IP connections, and deleted session files. On average, the automated analysis process completed within one minute per dataset, indicating a substantial reduction in investigatory time compared to manual inspection. These results collectively suggest that automated artifact extraction and correlation can enhance both the efficiency and consistency of mobile forensic investigations, particularly during the initial stages of incident analysis.

13.2. Comparison with Existing Forensic Tools

To better evaluate the effectiveness of the forensic analysis framework, a comparison with widely used mobile forensic tools was conducted. The selected tools include both commercial and open-source solutions commonly used in digital forensic investigations.

The comparison focuses on key criteria such as automation, artifact correlation, explainability, and support for application-level analysis.

Table 8 presents a summary of the comparison.

Table 8: Comparison of forensic tools

Tool	Automation	Artifact Correlation	Explainability	App-Level Analysis
Cellebrite UFED	Low	Limited	Medium	Low
Oxygen Forensic Detective	Medium	Limited	Medium	Medium
MobSF	Medium	No	High	High
Proposed System	High	Yes	High	High

The comparison shows that most existing tools focus primarily on data extraction rather than automated interpretation and correlation of artifacts. While commercial tools such as Cellebrite and Oxygen provide powerful acquisition capabilities, they often require significant manual analysis.

In contrast, the developed forensic platform emphasizes automation and artifact correlation, which significantly reduces analysis time and improves investigation efficiency. Another important aspect is, the system maintains high explainability due to its rule-based approach, making it suitable for forensic and legal contexts.

13.3. Advantages of the Proposed System

The implemented solution offers several advantages compared to existing solutions:

- Automated detection and correlation of forensic artifacts,
- High level of explainability due to rule-based analysis,
- Lightweight and modular architecture,
- Focus on application-level forensic analysis rather than full device extraction.

These features make the system particularly suitable for early-stage incident investigation and rapid analysis of desktop forensic application data.

14. Expected Outcomes and Contributions

The proposed research aims to deliver both practical and scientific contributions to the field of digital forensics, with a specific focus on forensic applications and security incident analysis.

14.1. Expected Outcomes

Compared to existing mobile forensic tools, the proposed system emphasizes application-level artifact analysis and automated correlation. While many tools focus on comprehensive data extraction, the proposed approach prioritizes interpretation and contextual linkage of artifacts related to security incidents.

This allows investigators to obtain faster insights, reduce manual analysis effort, and focus on high-risk indicators during the early stages of incident response.

The key outcomes of the project will include:

1. **Unified Forensic Framework:** Development of an integrated platform for mobile app forensic analysis that automates evidence acquisition, parsing, and correlation across memory, storage, and network layers.
2. **Mobile Artifact Taxonomy:** A comprehensive taxonomy describing the structure and semantics of mobile forensic artifacts across Android and iOS ecosystems. This taxonomy will serve as a shared reference for researchers and practitioners.
3. **Benchmark Datasets:** Construction of open datasets with verified forensic artifacts, supporting reproducible experiments and comparative evaluations of forensic tools.
4. **Prototype Implementation:** A functional proof-of-concept demonstrating how automation and modularity can accelerate digital investigations and improve consistency in forensic reporting.
5. **Practitioner Guidelines:** Documentation and best practices for investigators, ensuring standardization and legal defensibility of forensic procedures.

14.2. Scientific Contributions

- A systematic methodology for correlating heterogeneous mobile forensic artifacts to reconstruct temporal incident timelines.
- Quantitative evaluation of existing mobile forensic tools and artifact extraction accuracy.
- Formalization of a reproducible workflow for evidence validation and integrity verification.

14.3. Societal and Educational Impact

The system will enhance the efficiency and transparency of digital investigations. It can also serve as a training environment for forensic students and cybersecurity professionals, bridging the gap between theory and practical evidence-handling skills.

14.4. Potential Real-World Applications

The proposed forensic system can be applied in multiple real-world cybersecurity and digital investigation scenarios.

One of the primary applications is incident response, where investigators require rapid identification of suspicious artifacts and malicious activity indicators.

The system can also support malware investigations by identifying authentication tokens, suspicious network communication, and traces of unauthorized file manipulation.

In corporate environments, the proposed solution may assist security teams during internal security audits and desktop forensic application assessments.

Another important application area is cybersecurity education. The developed prototype can be used as a practical learning environment for students studying digital forensics, malware analysis, and incident response procedures.

Furthermore, the system may support preliminary forensic triage during large-scale investigations by automating repetitive analysis tasks and generating structured reports.

15. Limitations

Although the proposed framework seeks to be comprehensive, several limitations are acknowledged:

- **Platform Constraints:** Due to OS restrictions, certain low-level artifacts (e.g., encrypted memory regions or secure enclave data) may not be accessible without root or jailbreak privileges.
- **Dynamic Analysis Overhead:** Instrumentation and runtime monitoring may introduce latency or behavioral distortions in sensitive applications.
- **Data Volume and Storage:** Artifact extraction can generate large volumes of intermediary data, requiring scalable storage and filtering strategies.
- **Legal and Ethical Barriers:** Access to live systems or cloud-synced data may be subject to privacy laws and institutional ethics review.

Future improvements may include integration of machine learning techniques and support for real-time analysis.

15.1. Technical Limitations

Despite its advantages, the proposed system has several technical limitations. First, the reliance on rule-based detection restricts the ability to identify unknown or highly obfuscated threats. This may reduce effectiveness in complex or evolving attack scenarios.

Second, the system depends on access to application data, which may be limited by operating system restrictions or encryption mechanisms. In some cases, important artifacts may not be accessible without elevated privileges.

Third, the performance of the system may be affected by large datasets, as artifact extraction and analysis require processing significant volumes of data.

Addressing these limitations will be an important direction for future work.

15.2. Operational Constraints

In addition to technical limitations, several operational constraints may affect the practical deployment of the implemented solution.

One of the key challenges is the variability of desktop forensic application environments. Different operating system versions, device configurations, and application architectures may influence the availability and structure of forensic artifacts.

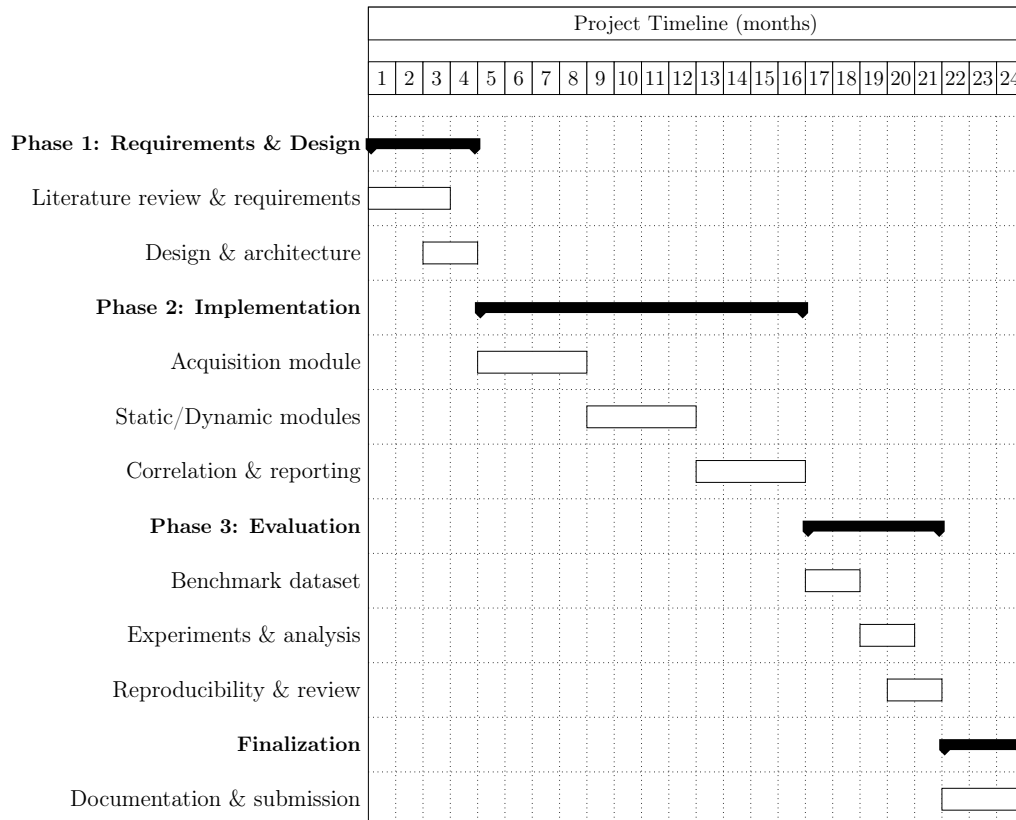
Another limitation is related to data accessibility. In real-world scenarios, access to application data may be restricted due to encryption, sandboxing mechanisms, or lack of user permissions. This can limit the effectiveness of artifact extraction and analysis.

Additionally, the system currently focuses on offline analysis of collected data. In time-critical scenarios, the lack of real-time monitoring capabilities may delay incident detection and response.

Finally, legal and organizational constraints may impact the use of forensic tools. Compliance with data protection regulations and internal policies must be ensured, which can restrict the scope of analysis.

Addressing these operational challenges will be essential for the successful adoption of the proposed system in real-world forensic investigations.

16. Timeline



17. Resources

The successful execution of this research will rely on a set of hardware, software, and data resources:

- **Hardware:** Virtualized Android and iOS devices, secure storage servers, and forensic workstations with high-memory configurations.
- **Software:** Open-source frameworks such as *MobSF*, *Wireshark*, *Autopsy*, *Volatility*, and custom Python-based parsers for application-specific data.
- **Datasets:** Public and synthetic forensic applications representing different threat categories (banking, social media, communications) will be used to simulate incident conditions.
- **Human Resources:** Collaboration with forensic experts, graduate researchers, and academic supervisors for evaluation and validation.

18. Ethical and Legal Considerations

The study will comply with established research ethics and data protection standards. All experiments will use consented or synthetic datasets that do not include personally identifiable information (PII). Evidence collection and handling will follow best practices of chain-of-custody, cryptographic hashing, and secure logging to ensure the authenticity and admissibility of evidence. Additionally, institutional approval will be obtained prior to any data acquisition involving live devices or applications.

19. Conclusion and Future Work

This research focused on the development of a Digital Forensic Examination System for forensic applications that bridges the gap between existing fragmented tools and the growing complexity of mobile ecosystems. The objectives defined at the beginning of this research were addressed during the development and evaluation of the proposed system.

The study included an analysis of existing forensic approaches and mobile forensic tools, identification of key forensic artifacts associated with desktop forensic analysis system environments, and the design of a modular forensic analysis architecture.

In addition, a Python-based forensic analysis prototype was implemented to perform artifact detection, risk evaluation, correlation, and automated report generation. Experimental testing demonstrated the ability of the system to identify suspicious artifacts and support structured forensic investigation procedures. By integrating artifact acquisition, dynamic analysis, correlation, and reporting, the developed framework was designed to improve investigation efficiency, reproducibility, and structured analysis procedures in digital investigations. The research objectives stated in Section 1.3 were addressed as follows:

1. **Analysis of existing tools** — completed in Sections 2 and 13; limitations of Cellebrite, Oxygen, and MobSF were identified (Table 6).
2. **Identification of key forensic artifacts** — a taxonomy of artifacts was proposed in Section 3 (Table 2), covering authentication tokens, network indicators, file deletion traces, and session records.
3. **System architecture design** — a modular desktop architecture was designed in Section 5 (Figure 1).
4. **Development of the forensic prototype** — a functional Python desktop application was implemented in Section 7, including GUI, scanner, detection engine, and report generator.
5. **Anomaly detection implementation** — rule-based detection with risk scoring was implemented and described in Sections 7.3–7.4.
6. **System evaluation** — experimental evaluation on four datasets yielded Precision = 0.875, Recall = 0.808, Accuracy = 0.949, confirming effectiveness of the approach (Section 8).

Future work will explore:

- Integration of cloud and IoT artifact correlation for distributed incident reconstruction;
- Real-time forensic data collection using trusted execution environments (TEEs);
- AI-assisted artifact classification and anomaly detection;
- Development of community-driven, open-source forensic datasets and validation benchmarks.

Overall, the developed solution provides a robust foundation for advancing digital forensics research and establishing standardized methodologies for mobile application analysis.

19.1. Final Remarks

This research demonstrates that automated forensic analysis of mobile applications is both feasible and effective when based on a structured and explainable methodology.

The developed forensic platform integrates artifact detection, correlation, and risk evaluation into a unified analysis workflow. Artifact detection, correlation, and risk evaluation into a unified workflow. Unlike traditional approaches that rely heavily on manual analysis, the system reduces investigation time while maintaining transparency and reproducibility.

The experimental evaluation and case study suggest that the system is capable of detecting key forensic artifacts and providing meaningful investigation insights. That the system can reliably detect key forensic artifacts and provide meaningful insights into potential security incidents. The integration of a risk scoring model further enhances the ability to prioritize findings and support decision-making processes.

Overall, this work contributes to the advancement of mobile digital forensics by emphasizing application-level analysis and automation. It provides a practical foundation for future research and development in this area.

19.2. Future Research Directions

Although the implemented solution demonstrated generally stable performance in detecting predefined forensic artifacts, several directions for future research can be identified.

First, the integration of machine learning techniques could significantly enhance the system's ability to detect unknown and complex patterns. Methods such as anomaly detection and classification models may improve accuracy and reduce false positives.

Second, the system can be extended to support real-time forensic analysis. This would enable continuous monitoring of mobile applications and faster detection of security incidents as they occur.

Third, future work may include integration with cloud-based forensic platforms. As mobile applications increasingly rely on cloud services, the ability to correlate local and remote artifacts will become essential for comprehensive incident analysis.

Finally, the development of standardized benchmark datasets would support more rigorous evaluation and comparison of forensic tools, contributing to the advancement of the field.

These research directions demonstrate that the proposed system can be further expanded and adapted for more complex forensic investigation scenarios, for further development of the proposed system and its applicability in real-world forensic investigations.

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A. Appendix: Source Code

This appendix presents selected fragments of the implemented Python source code used in the developed forensic examination system.

A.1. Artifact Detection Logic

```
def analyze(content):
    findings = []

    if "token=" in content:
        findings.append(("Token", "CRITICAL"))

    if re.search(r'\d+\. \d+\. \d+\. \d+', content):
        findings.append(("Network", "HIGH"))

    if "delete" in content.lower():
        findings.append(("File deletion", "MEDIUM"))

    if "auth" in content.lower():
        findings.append(("Authentication", "LOW"))

    return findings
```

A.2. Risk Scoring Module

```
def calculate_score(findings):
    score = 0

    for f in findings:
        if f[1] == "CRITICAL":
            score += 5
        elif f[1] == "HIGH":
            score += 3
        elif f[1] == "MEDIUM":
            score += 2
        elif f[1] == "LOW":
            score += 1

    return score
```

A.3. AI-based Analysis Module

```
def ai_analysis(findings):  
    score = calculate_score(findings)  
  
    if score >= 8:  
        return "High likelihood of compromise detected."  
    elif score >= 5:  
        return "Moderate risk detected."  
    else:  
        return "Low risk."
```

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